Jeroen Van Goey

Staff Research Engineer in BioAI

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Summary

Machine learning research engineer working at the intersection of AI and science, bridging scientific ML, biological domain knowledge, production-quality software, and team leadership. I build generative models and take them from prototype to deployment, with a focus on *de novo* peptide sequencing.

Author of 8 publications, including in *Nature Machine Intelligence*.

Currently looking for Staff/Principal Research Engineer roles in AI for science. Based in Cape Town, open to remote-first roles now and hybrid/on-site roles in Europe from 2027.

Experience

InstaDeep, Staff Research Engineer in BioAI — Cape Town, South Africa
Aug 2022 – present
3 years 11 months

- **Tech lead and hiring manager** for two cross-disciplinary **ML research teams** of about 5 to 8 people: *de novo* peptide sequencing (with the Technical University of Denmark) and signal-peptide design for secretion efficiency (with BioNTech).
- Delivered a **client collaboration with Syngenta**, applying **genomic language models** (AgroNT, a Nucleotide Transformer trained on ~10.5 million genomic sequences spanning trillions of base pairs across 48 plant species) to accelerate crop trait research.
- Set **research direction** and take models **from research to production**, balancing scientific rigour with reliable, maintainable engineering.
- **Lead the BioAI department** of InstaDeep's Cape Town office and serve as the office's **site manager** (an office of about 25 people); hire and grow the BioAI teams across the Cape Town and Kigali offices.
- Directed the team's **ML engineering foundations**: scalable, Transformer-based libraries (Python / PyTorch) for large-scale training on InstaDeep's Kyber (~500 PFLOPs) and EuroHPC's MareNostrum 5 (260 PFLOPs) supercomputers, and the cloud (AWS, GCP).

Barco, Senior Software Development Engineer, Machine Learning — Kortrijk, Belgium
Feb 2020 – Aug 2022
2 years 7 months

- Built a TensorFlow Extended production pipeline (orchestrated with Apache Airflow) training deep-learning models on multispectral images to detect and classify melanoma skin cancers for Demetra, Barco's dermatology imaging device.
- Engineered the pipeline with **end-to-end data and model lineage tracking** to deliver the traceability and reproducibility that **regulated medical industries (FDA approval)** require.
- Worked across the research, cloud-backend and mobile/web frontend teams.

Bayer Crop Science, then BASF, Bioinformatics Researcher, Manager of the Python & R Platforms — Zwijnaarde, Belgium
Feb 2018 – Jan 2020
2 years

- Owned the **Python/R data-analysis platform** used by **480 researchers and data scientists**: technical support, proactive monitoring and root-cause analysis, **training** (NumPy, pandas, BioPython) and **mentoring** across the company's internal research community.
- The Zwijnaarde site was divested from Bayer Crop Science to BASF as part of Bayer's acquisition of Monsanto; I continued in the same role across the transition.

Applied Maths (acquired by bioMérieux), Bioinformatics Software Developer — Sint-Martens-Latem, Belgium
Sept 2011 – Jan 2018
6 years 5 months

- Worked on BioNumerics, a bioinformatics suite for integrated analysis of biological data, writing **custom Python scripts and extensions** for clients in the clinical, academic and government sectors.

Selected publications

InstaNovo enables diffusion-powered *de novo* peptide sequencing in large-scale proteomics experiments — Nature Machine Intelligence, 2025
K. Eloff, K. Kalogeropoulos, …, **J. Van Goey**, …, T. P. Jenkins
Trained on ~63 million spectra; top-ranked in an independent benchmark of 17 de novo tools across 83 datasets; Altmetric attention score 100+ and 120+ GitHub stars.
10.1038/s42256-025-01019-5

InstaNovo-P: a *de novo* peptide sequencing model for phosphoproteomics — bioRxiv, 2025
J. Lauridsen, P. Ramasamy, …, **J. Van Goey**, …, K. Kalogeropoulos
10.1101/2025.05.14.654049

***De novo* peptide sequencing rescoring and FDR estimation with Winnow** — arXiv, 2025
A. Mabona, J. Daniel, …, **J. Van Goey**, K. Kalogeropoulos
10.48550/arXiv.2509.24952

A living proteomics benchmark for comprehensive evaluation of deep learning-based *de novo* peptide sequencing tools — Registered report (preprint), 2026
M. Pominova, **J. Van Goey**, C. Adams, …, W. Bittremieux
10.6084/m9.figshare.31680883.v1

Generalizable direct protein sequencing with InstaNexus — Mol. Cell. Proteomics, 2026
M. Reverenna, M. Wennekers Nielsen, …, **J. Van Goey**, …, K. Kalogeropoulos
10.1016/j.mcpro.2026.101547

A Framework for Database Search with AI Models in Mass Spectrometry-Based Proteomics — Journal of Proteome Research, 2026
K. Kalogeropoulos, **J. Van Goey**, T. P. Jenkins, K. M. Eloff
10.1021/acs.jproteome.5c01153

Education

Microdegree — **Hogeschool West-Vlaanderen (Howest)**, Machine Learning & Deep Learning — Belgium, 2018 – 2019

Katholieke Universiteit Leuven, Bioinformatics — Leuven, Belgium, 2010 – 2012
Partial credits obtained.

M.Sc. — **University of Antwerp**, Biology — Antwerp, Belgium, 1997 – 2004

Skills

Machine learning: Transformers, diffusion models, large language models (LLMs), supervised & self-supervised learning
Frameworks: PyTorch, TensorFlow / TFX, NumPy, pandas
Scale & MLOps: Distributed GPU training, HPC (InstaDeep's Kyber and EuroHPC's MareNostrum 5 supercomputers), Kubernetes-based ML platforms (AIchor), cloud (AWS, GCP), Docker, CI/CD, workflow orchestration (Airflow, Dagster), experiment tracking (MLflow, Neptune, TensorBoard)
Bioinformatics: BioPython, BLAST, ClustalW, Snakemake, BioNumerics
Languages: Python (primary), R

Projects

Awesome De Novo Peptide Sequencing
A comprehensive, interactive map of the *de novo* peptide sequencing field: algorithms, post-processors, downstream applications, and adjacent tools, deep-learning and classical alike.
- Open source: github.com/BioGeek/awesome_de_novo_peptide_sequencing

Teaching & mentoring

Machine Learning for Biology practical — Deep Learning Indaba, Ghana, 2023
Mentored the hands-on practical alongside InstaDeep and Google DeepMind colleagues.

Scientific Python training & mentoring — BASF and Bayer Crop Science, Belgium, 2018 – 2020
Trained and mentored researchers and data scientists in scientific Python (NumPy, pandas, BioPython) as manager of the Python/R analysis platform.

Hackathons (co-created, organized and judged)

2025: Machine Learning Interatomic Potentials Hackathon, IndabaX (Stellenbosch University, South Africa)
2024: Snakes and Sequences: Senegalese Serpent Venom Sequencing Hackathon, Deep Learning Indaba (Senegal)
2024: Desert Locust Breeding Ground Prediction Hackathon, IndabaX (Wits, Johannesburg)
2023: Unveiling Cassava's Secrets, Deep Learning Indaba (Ghana)